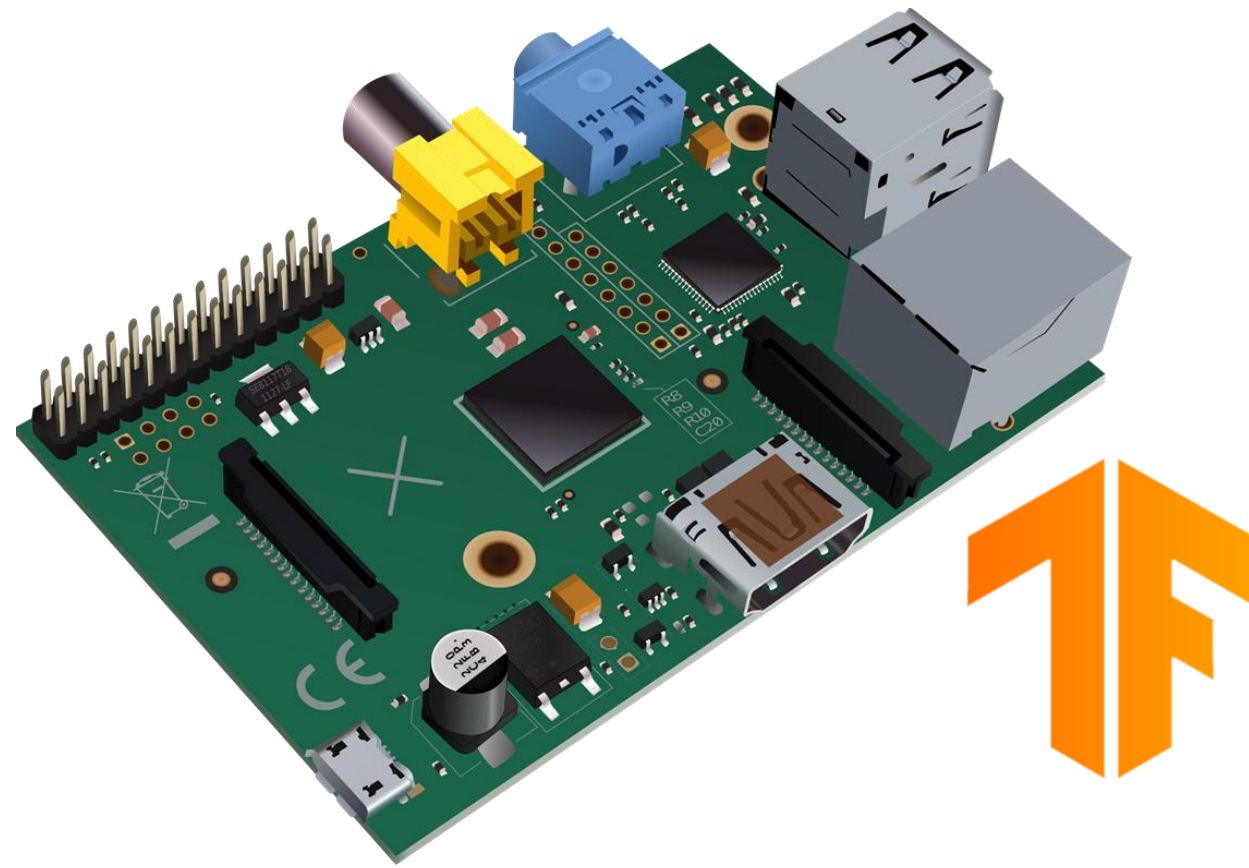
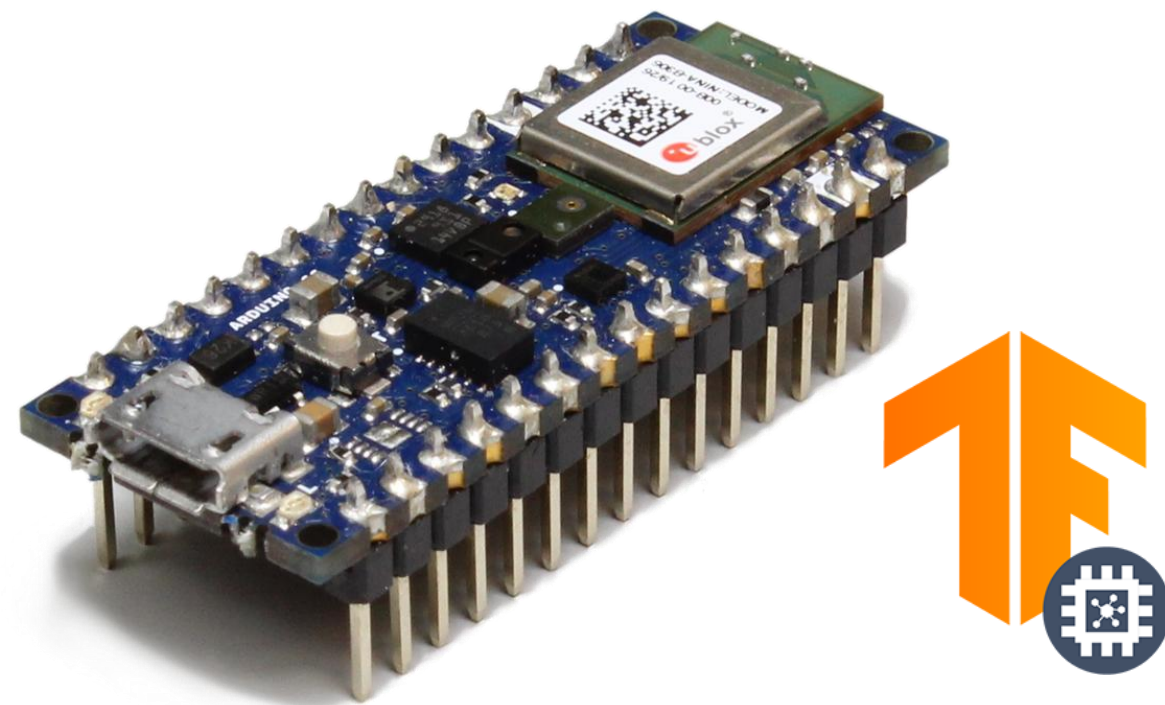




Single Board Computer

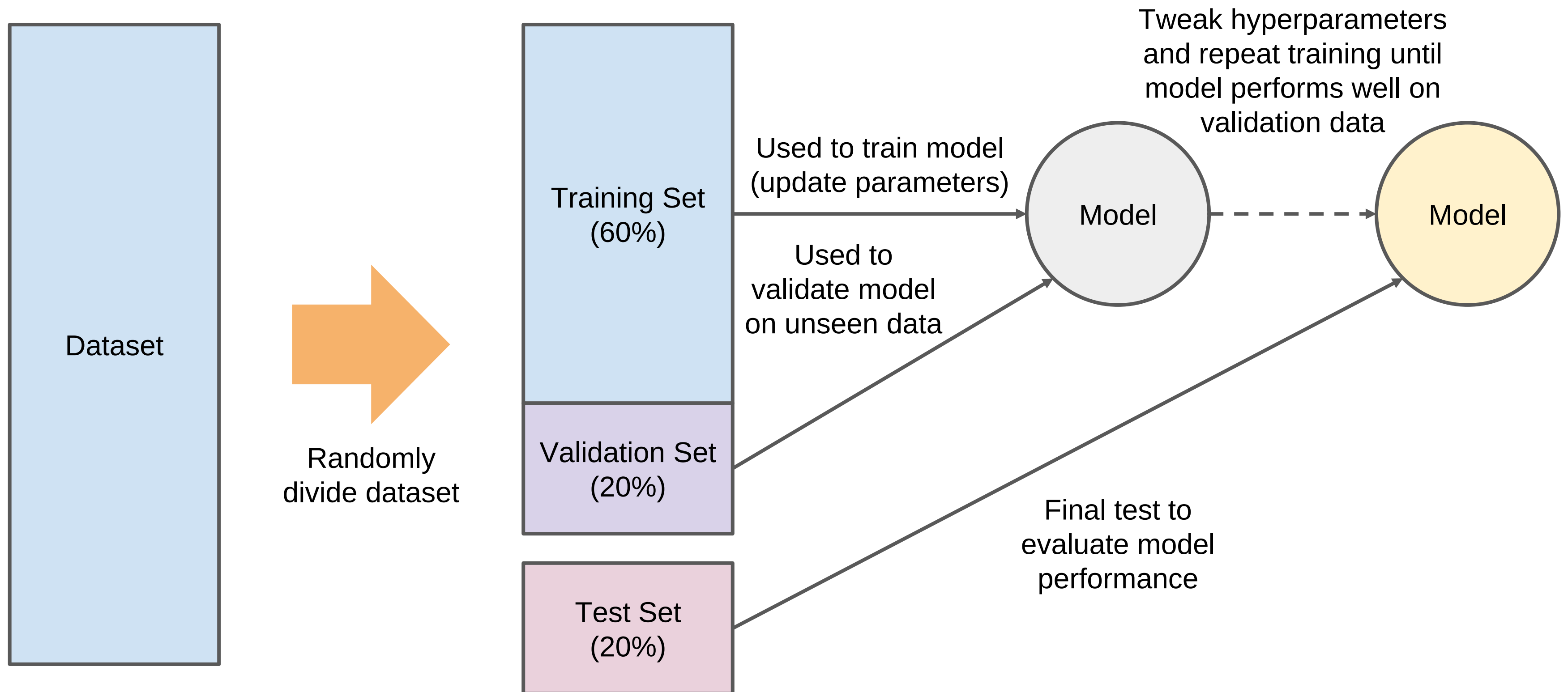
- More powerful (faster processor, more memory)
- Runs full, general purpose operating system (OS)
- Can provide full command line or graphical user interface
- Requires more power



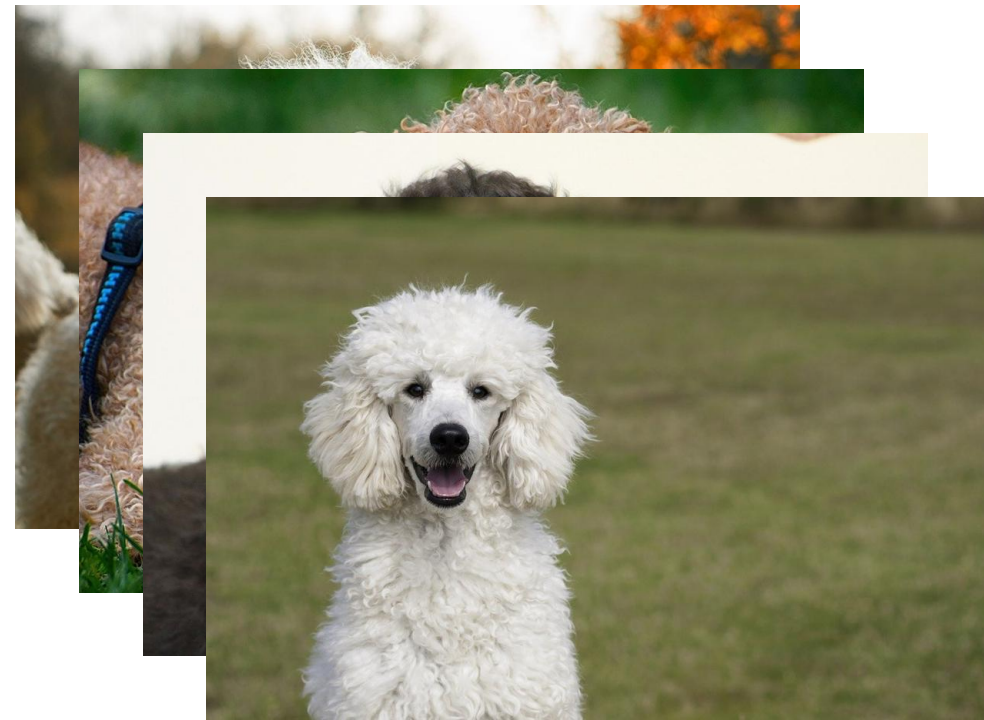
Microcontroller

- Less powerful
- Bare-metal (superloop) or real-time operating system (RTOS)
- Limited or no user interface
- Requires less power

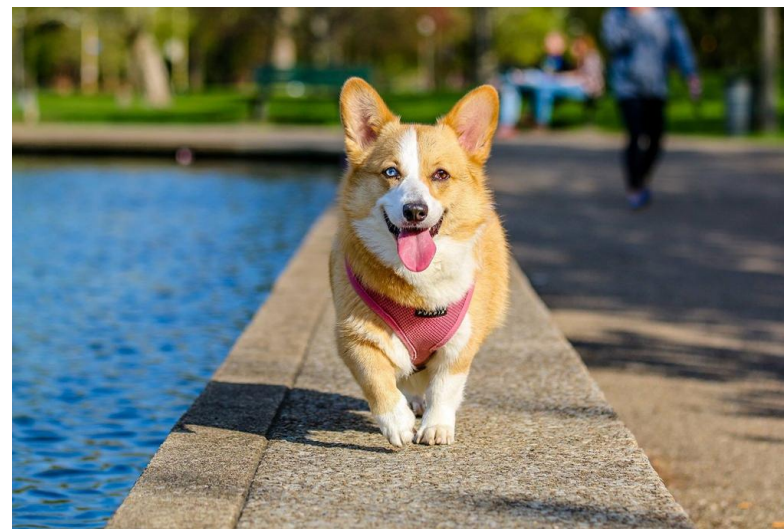
Holdout Method



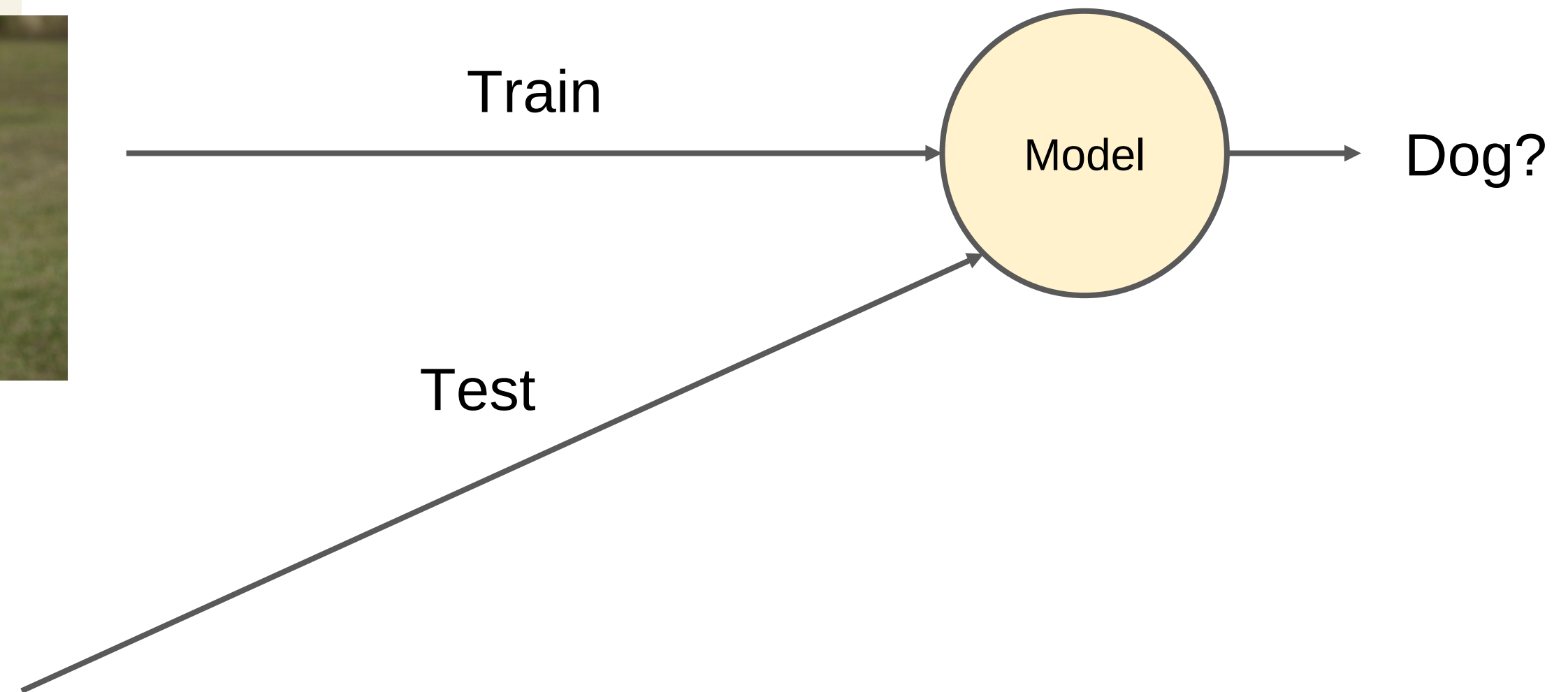
Think about your data!



Training Data



Test Data



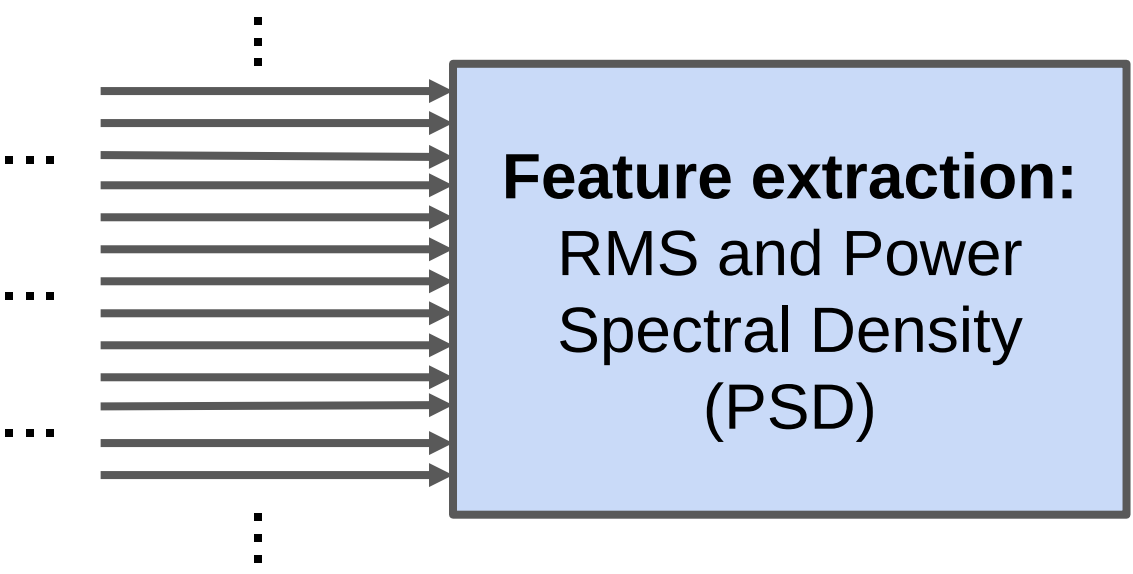
Feature Example



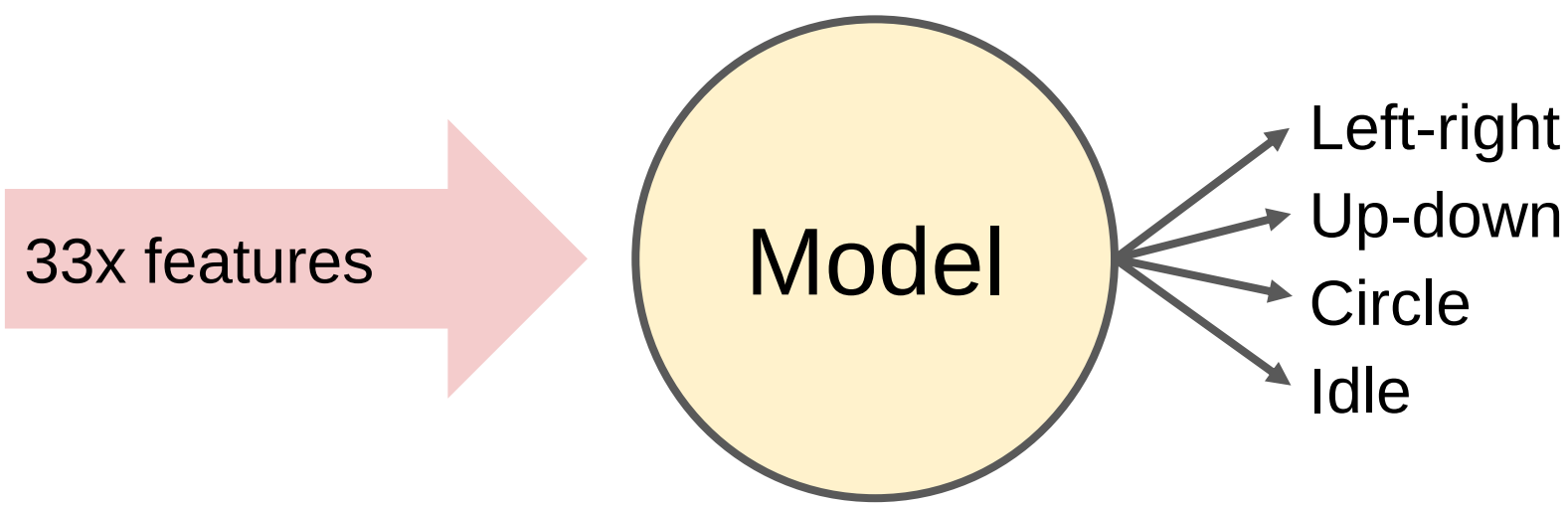
11 features per axis: RMS, 3x peak amplitudes from PSD, 3x peak frequencies from PSD, 4x spectral bins

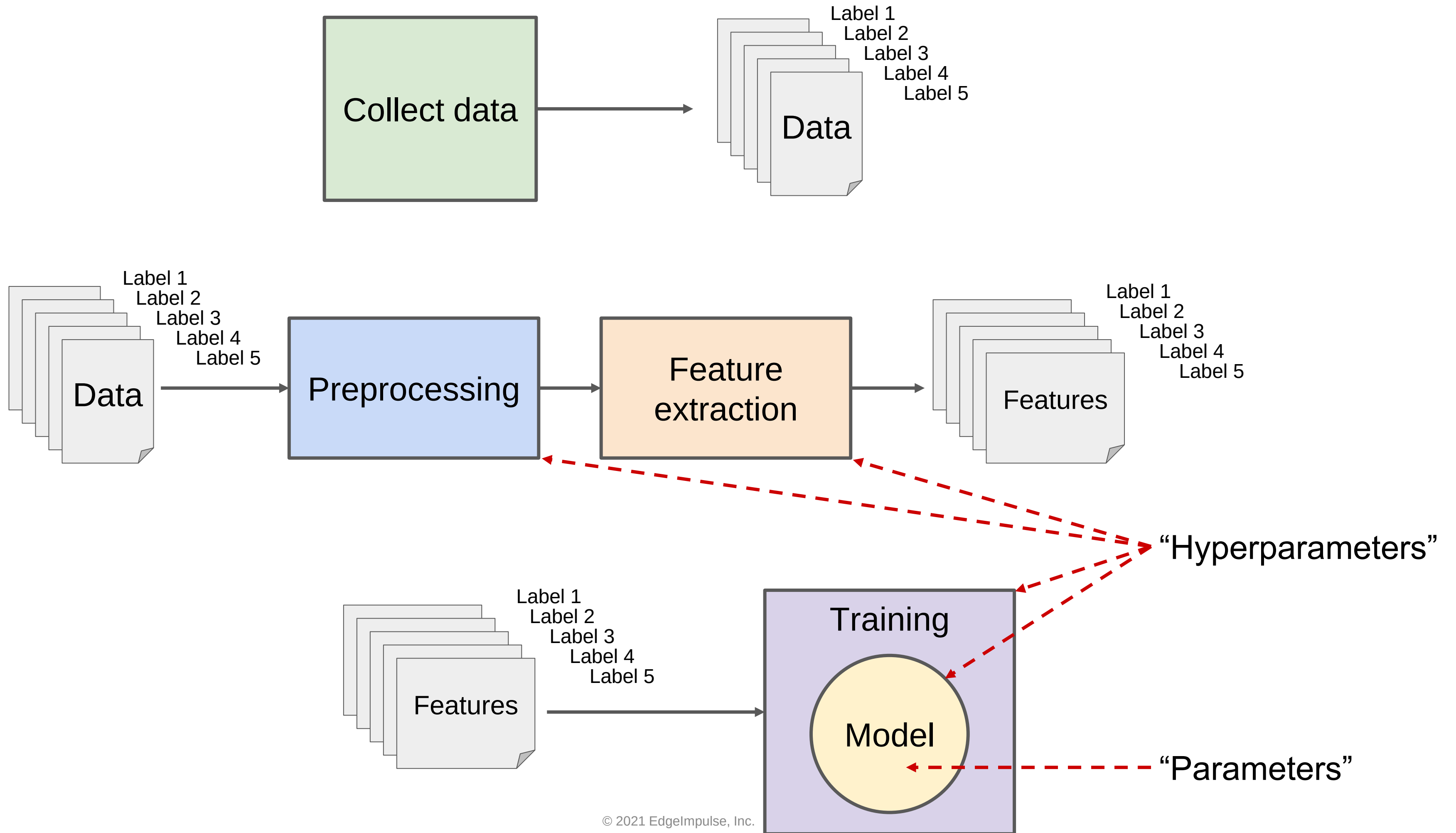
Acceleration (m/s^2)

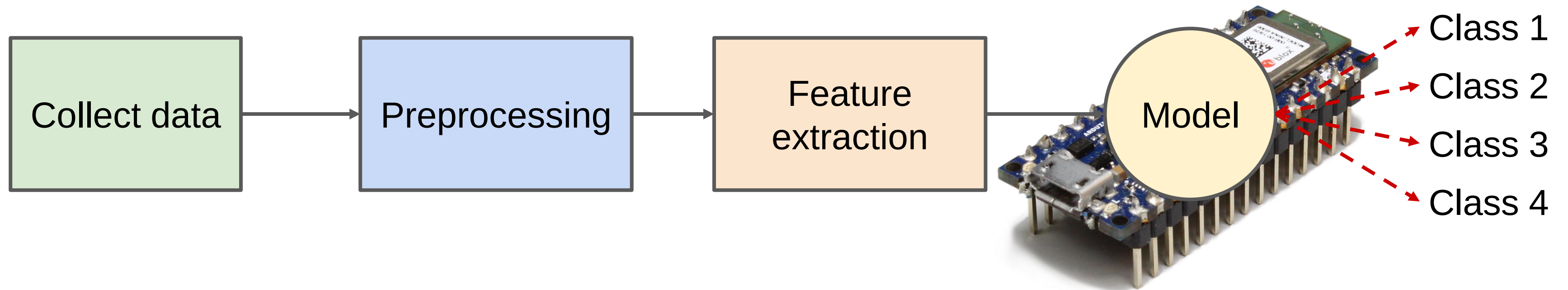
x	-6.8	-5.6	-4.6	-4.1	-3.8
y	-0.1	0.2	0.7	0.9	0.7
z	10.4	10.3	10.2	10.1	9.6



375 raw values







Inference: using the machine learning model to make predictions on unseen data in the wild